

CAPE PHILLIPS, Antarctica

WMO: 896610

Lat: 73.518S

Lon: 169.749E

Elev: 569

StdP: 94.67

Time Zone: 12.00 (NZL)

Period: 01-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	-34.2	-32.8	N/A	N/A	N/A	N/A	N/A	N/A	37.7	-24.8	35.1	-21.4	6.3	240	1.444

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	2.8	0.1	N/A	-1.6	N/A	-3.0	N/A	N/A	N/A	N/A	N/A	N/A	N/A	3.6	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 35.3	30.4	26.3	DB	-36.1	3.1	1.0	1.9	-36.8	4.5	-37.4	5.6	-37.9	6.6	-38.6	8.0	(4)
(5)			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	-16.9	-5.4	-9.5	-15.5	-20.8	-21.3	-23.3	-24.5	-22.9	-22.2	-19.0	-11.5	-6.2	(6)	
	DBStd	7.83	1.79	2.37	3.47	4.48	5.20	5.02	4.90	5.09	4.80	3.43	3.81	2.31	(7)	
	HDD10.0	9809	478	547	789	923	969	1000	1071	1022	966	900	644	501	(8)	
	HDD18.3	12851	737	780	1048	1173	1227	1250	1329	1280	1216	1158	894	759	(9)	
	CDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0	(10)	
	CDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0	(11)	
	CDH23.3	0	0	0	0	0	0	0	0	0	0	0	0	0	(12)	
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0	(13)	
(14) Wind	WSAvg	8.7	8.0	8.3	8.7	7.9	8.8	9.8	9.3	9.9	9.5	10.2	7.9	6.0	(14)	
(15) Precipitation	PrecAvg	364	42	48	37	25	28	27	23	27	27	27	26	28	(15)	
	PrecMax	497	81	116	71	53	59	53	53	60	58	55	59	54	(16)	
	PrecMin	220	13	19	21	7	5	6	7	0	0	10	8	7	(17)	
	PrecStd	62	18	20	12	12	15	11	11	14	16	12	16	13	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	2.0	-2.3	-8.1	-7.2	-6.4	-9.9	-11.6	-5.8	-8.7	-7.6	-0.5	3.1	(19)	
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(20)	
	2%	DB	-0.3	-4.6	-9.0	-10.8	-10.3	-12.8	-13.8	-11.0	-11.5	-10.6	-2.6	0.5	(21)	
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(22)	
	5%	DB	-1.7	-5.5	-9.9	-12.7	-12.7	-14.5	-16.3	-14.1	-13.3	-12.9	-5.0	-1.4	(23)	
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(24)	
	10%	DB	-2.8	-6.4	-11.0	-15.2	-14.5	-16.1	-16.7	-16.2	-15.3	-14.3	-6.6	-3.1	(25)	
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(27)	
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(28)	
	2%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(29)	
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(30)	
	5%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(31)	
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(32)	
	10%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(33)	
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(34)	
(35) Mean Daily Temperature Range		MDBR	2.8	2.7	3.0	4.1	4.8	5.0	4.8	5.7	5.1	3.7	3.8	3.4	(35)	
	5% DB	MCDBR	5.1	4.0	4.2	6.8	8.1	6.7	6.4	9.7	8.5	5.9	5.3	5.7	(36)	
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(37)	
	5% WB	MCDBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(38)	
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(39)	
(40) Clear-Sky Solar Irradiance	taub	0.245	0.238	0.197	0.110	N/A	N/A	N/A	N/A	0.108	0.174	0.197	0.218	0.237	(40)	
	taud	2.427	2.422	2.333	1.922	N/A	N/A	N/A	N/A	1.901	2.250	2.334	2.343	2.366	(41)	
	Ebn at Noon	989	925	837	545	0	0	0	0	528	856	975	1018	1017	(42)	
	Edh at Noon	89	74	53	20	0	0	0	0	19	55	79	96	99	(43)	
(44) All-Sky Solar Radiation	RadAvg	6.21	3.26	1.57	0.36	0.00	0.00	0.00	0.00	0.12	1.39	4.03	7.17	8.36	(44)	
	RadStd	0.69	0.44	0.24	0.02	0.00	0.00	0.00	0.00	0.01	0.08	0.15	0.27	0.53	(45)	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (1 neighbor)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page